2 Subject content

Qualification aims and objectives

GCSE study in the sciences provides the foundation for understanding the material world. Scientific understanding is changing our lives and is vital to the world's future prosperity. All students should learn essential aspects of the knowledge, methods, processes and uses of science. They should gain appreciation of how the complex and diverse phenomena of the natural world can be described in terms of a small number of key ideas that relate to the sciences and that are both inter-linked and of universal application. These key ideas include:

- the use of conceptual models and theories to make sense of the observed diversity of natural phenomena
- the assumption that every effect has one or more cause
- that change is driven by differences between different objects and systems when they interact
- that many such interactions occur over a distance without direct contact
- that science progresses through a cycle of hypothesis, practical experimentation, observation, theory development and review
- that quantitative analysis is a central element both of many theories and of scientific methods of inquiry.

These key ideas are relevant in different ways and with different emphases in the three subjects. Examples of their relevance are given for each subject in the separate sections below for Biology, Chemistry and Physics.

The three GCSE Science qualifications enable students to:

- develop scientific knowledge and conceptual understanding through the specific disciplines of Biology, Chemistry and Physics
- develop understanding of the nature, processes and methods of science, through different types of scientific enquiries that help them to answer scientific questions about the world around them
- develop and learn to apply observational, practical, modelling, enquiry and problem-solving skills in the laboratory, in the field and in other learning environments
- develop their ability to evaluate claims based on science through critical analysis of the methodology, evidence and conclusions, both qualitatively and quantitatively.

Students should study the sciences in ways that help them to develop curiosity about the natural world, that give them an insight into how science works and that enable them to appreciate its relevance to their everyday lives. The scope and nature of the study should be broad, coherent, practical and satisfying. It should encourage students to be inspired, motivated and challenged by the subject and its achievements.

The key ideas specific to the Chemistry content include:

- matter is composed of tiny particles called atoms and there are about 100 different naturally occurring types of atoms called elements
- elements show periodic relationships in their chemical and physical properties

- the shapes of molecules (groups of atoms bonded together) and the way giant structures are arranged is of great importance in terms of the way they behave
- there are barriers to reaction so reactions occur at different rates
- chemical reactions take place in only three different ways:
  - proton transfer
  - electron transfer
  - electron sharing
- energy is conserved in chemical reactions so can therefore be neither created nor destroyed.

All of these key ideas will be assessed as part of this qualification, through the subject content.

Working scientifically

The GCSE in Chemistry requires students to develop the skills, knowledge and understanding of working scientifically. Working scientifically will be assessed through examination and the completion of the eight core practicals.

1 Development of scientific thinking

a Understand how scientific methods and theories develop over time.

b Use a variety of models, such as representational, spatial, descriptive, computational and mathematical, to solve problems, make predictions and to develop scientific explanations and an understanding of familiar and unfamiliar facts.

c Appreciate the power and limitations of science, and consider any ethical issues that may arise.

d Explain everyday and technological applications of science; evaluate associated personal, social, economic and environmental implications; and make decisions based on the evaluation of evidence and arguments.

e Evaluate risks both in practical science and the wider societal context, including perception of risk in relation to data and consequences.

f Recognise the importance of peer review of results and of communicating results to a range of audiences.

2 Experimental skills and strategies

a Use scientific theories and explanations to develop hypotheses.

b Plan experiments or devise procedures to make observations, produce or characterise a substance, test hypotheses, check data or explore phenomena.

c Apply a knowledge of a range of techniques, instruments, apparatus and materials to select those appropriate to the experiment.

d Carry out experiments appropriately, having due regard to the correct manipulation of apparatus, the accuracy of measurements and health and safety considerations.

e Recognise when to apply a knowledge of sampling techniques to ensure any samples collected are representative.

f Make and record observations and measurements using a range of apparatus and methods.

g Evaluate methods and suggest possible improvements and further investigations.

3 Analysis and evaluation

Apply the cycle of collecting, presenting and analysing data, including:

a presenting observations and other data using appropriate methods.

b translating data from one form to another.

c carrying out and representing mathematical and statistical analysis.

d representing distributions of results and making estimations of uncertainty.

e interpreting observations and other data (presented in verbal, diagrammatic, graphical, symbolic or numerical form), including identifying patterns and trends, making inferences and drawing conclusions.

f presenting reasoned explanations, including relating data to hypotheses.

g being objective, evaluating data in terms of accuracy, precision, repeatability and reproducibility and identifying potential sources of random and systematic error.

h communicating the scientific rationale for investigations, methods used, findings and reasoned conclusions through paper-based and electronic reports and presentations using verbal, diagrammatic, graphical, numerical and symbolic forms.

4 Scientific vocabulary, quantities, units, symbols and nomenclature

a Use scientific vocabulary, terminology and definitions.

b Recognise the importance of scientific quantities and understand how they are determined.

c Use SI units (e.g. kg, g, mg; km, m, mm; kJ, J) and IUPAC chemical nomenclature unless inappropriate.

d Use prefixes and powers of ten for orders of magnitude (e.g. tera, giga, mega, kilo, centi, milli, micro and nano).

e Interconvert units.

f Use an appropriate number of significant figures in calculation.

Practical work

The content includes eight mandatory core practicals, indicated as an entire specification point in italics.

Students must carry out all eight of the mandatory core practicals listed below.

Core practical:

2.11 *Investigate the composition of inks using simple distillation and paper chromatography*

3.6 *Investigate the change in pH on adding powdered calcium hydroxide or calcium oxide to a fixed volume of dilute hydrochloric acid*

3.17 *Investigate the preparation of pure, dry hydrated copper sulfate crystals starting from copper oxide including the use of a water bath*

3.31 *Investigate the electrolysis of copper sulfate solution with inert electrodes and copper electrodes*

5.9C *Carry out an accurate acid-alkali titration, using burette, pipette and a suitable indicator*

7.1 *Investigate the effects of changing the conditions of a reaction on the rates of chemical reactions by:*

a measuring the production of a gas (in the reaction between hydrochloric acid and marble chips)

b observing a colour change (in the reaction between sodium thiosulfate and hydrochloric acid)

9.6C *Identify the ions in unknown salts, using the tests for the specified cations and anions in 9.2C, 9.3C, 9.4C, 9.5C*

9.28C *Investigate the temperature rise produced in a known mass of water by the combustion of the alcohols ethanol, propanol, butanol and pentanol*

Students will need to use their knowledge and understanding of these practical techniques and procedures in the written assessments.

Centres must confirm that each student has completed the eight mandatory core practicals.

Students need to record the work that they have undertaken for the eight mandatory core practicals. The practical record must include the knowledge, skills and understanding they have derived from the practical activities. Centres must complete and submit a Practical Science Statement (see *Appendix 5*) to confirm that all students have completed the eight mandatory core practicals. This must be submitted to Pearson by 15th April in the year that the students will sit their examinations. Any failure by centres to provide this Practical Science Statement will be treated as malpractice and/or maladministration.

Scientific diagrams should be included, where appropriate, to show the set-up and to record the apparatus and procedures used in practical work.

It is important to realise that these core practicals are the minimum number of practicals that should be taken during the course. Suggested additional practicals are given beneath the content at the end of each topic. The eight mandatory core practicals cover all aspects of the apparatus and techniques listed in *Appendix 4: Apparatus and techniques*. This appendix also includes more detailed instructions for each core practical, which must be followed.

Safety is an overriding requirement for all practical work. Centres are responsible for ensuring appropriate safety procedures are followed whenever their students complete practical work.

These core practicals may be reviewed and amended if changes are required to the apparatus and techniques listed by the Department for Education. Pearson may also review and amend the core practicals if necessary. Centres will be told as soon as possible about any changes to core practicals.

Qualification content

The following notation is used in the tables that show the content for this qualification:

- text in **bold** indicates content that is for higher tier only
- entire specification points in italics indicates a core practical.

Specification statement numbers with a C in them refer to content which is only in the GCSE in Chemistry and is not found in the GCSE in Combined Science (e.g. 5.1C).

Mathematics

Maths skills that can be assessed in relation to a specification point are referenced in the maths column, next to this specification point. Please see *Appendix 1: Mathematical skills* for full details of each maths skill.

After each topic of content in this specification, there are details relating to the 'Use of mathematics' which contains the Chemistry specific mathematic skills that are found in each topic of content in the document *Biology, Chemistry and Physics GCSE subject content*, published by the Department for Education (DfE) in June 2014. The reference in brackets after each statement refers to the mathematical skills from *Appendix 1*.

Topics common to Paper 1 and Paper 2

Formulae, equations and hazards

Students should:		Maths skills
0.1	Recall the formulae of elements, simple compounds and ions	
0.2	Write word equations	
0.3	Write balanced chemical equations, including the use of the state symbols (s), (l), (g) and (aq)	1c
0.4	**Write balanced ionic equations**	1c
0.5	Describe the use of hazard symbols on containers a to indicate the dangers associated with the contents b to inform people about safe-working precautions with these substances in the laboratory	
0.6	Evaluate the risks in a practical procedure and suggest suitable precautions for a range of practicals including those mentioned in the specification	

Use of mathematics

- Arithmetic computation and ratio when balancing equations (1a and 1c).

Topic 1 – Key concepts in chemistry

Atomic structure

Students should:		Maths skills
1.1	Describe how the Dalton model of an atom has changed over time because of the discovery of subatomic particles	
1.2	Describe the structure of an atom as a nucleus containing protons and neutrons, surrounded by electrons in shells	
1.3	Recall the relative charge and relative mass of: a a proton b a neutron c an electron	
1.4	Explain why atoms contain equal numbers of protons and electrons	
1.5	Describe the nucleus of an atom as very small compared to the overall size of the atom	1d
1.6	Recall that most of the mass of an atom is concentrated in the nucleus	
1.7	Recall the meaning of the term mass number of an atom	
1.8	Describe atoms of a given element as having the same number of protons in the nucleus and that this number is unique to that element	
1.9	Describe isotopes as different atoms of the same element containing the same number of protons but different numbers of neutrons in their nuclei	
1.10	Calculate the numbers of protons, neutrons and electrons in atoms given the atomic number and mass number	3b
1.11	Explain how the existence of isotopes results in relative atomic masses of some elements not being whole numbers	1a, 1c
1.12	**Calculate the relative atomic mass of an element from the relative masses and abundances of its isotopes**	1a, 1c, 1d 3a, 3c

Use of mathematics

- Relate size and scale of atoms to objects in the physical world (1d).
- Estimate size and scale of atoms (1d).

The periodic table

Students should:		Maths skills
1.13	Describe how Dmitri Mendeleev arranged the elements, known at that time, in a periodic table by using properties of these elements and their compounds	
1.14	Describe how Dmitri Mendeleev used his table to predict the existence and properties of some elements not then discovered	
1.15	Explain that Dmitri Mendeleev thought he had arranged elements in order of increasing relative atomic mass but this was not always true because of the relative abundance of isotopes of some pairs of elements in the periodic table	
1.16	Explain the meaning of atomic number of an element in terms of position in the periodic table and number of protons in the nucleus	
1.17	Describe that in the periodic table a elements are arranged in order of increasing atomic number, in rows called periods b elements with similar properties are placed in the same vertical columns called groups	
1.18	Identify elements as metals or non-metals according to their position in the periodic table, explaining this division in terms of the atomic structures of the elements	
1.19	Predict the electronic configurations of the first 20 elements in the periodic table as diagrams and in the form, for example 2.8.1	4a 5b
1.20	Explain how the electronic configuration of an element is related to its position in the periodic table	4a

Ionic bonding

Students should:		Maths skills
1.21	Explain how ionic bonds are formed by the transfer of electrons between atoms to produce cations and anions, including the use of dot and cross diagrams	5b
1.22	Recall that an ion is an atom or group of atoms with a positive or negative charge	
1.23	Calculate the numbers of protons, neutrons and electrons in simple ions given the atomic number and mass number	3b
1.24	Explain the formation of ions in ionic compounds from their atoms, limited to compounds of elements in groups 1, 2, 6 and 7	1c 5b
1.25	Explain the use of the endings –ide and –ate in the names of compounds	

Students should:		Maths skills
1.26	Deduce the formulae of ionic compounds (including oxides, hydroxides, halides, nitrates, carbonates and sulfates) given the formulae of the constituent ions	1c
1.27	Explain the structure of an ionic compound as a lattice structure a consisting of a regular arrangement of ions b held together by strong electrostatic forces (ionic bonds) between oppositely-charged ions	5b

Use of mathematics

- Represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures (5b).

Covalent bonding

Students should:		Maths skills
1.28	Explain how a covalent bond is formed when a pair of electrons is shared between two atoms	
1.29	Recall that covalent bonding results in the formation of molecules	
1.30	Recall the typical size (order of magnitude) of atoms and small molecules	1d
1.31	Explain the formation of simple molecular, covalent substances, using dot and cross diagrams, including: a hydrogen b hydrogen chloride c water d methane e oxygen f carbon dioxide	5b

Use of mathematics

- Relate size and scale of atoms to objects in the physical world (1d).
- Estimate size and scale of atoms (1d).

Types of substance

Students should:		Maths skills
1.32	Explain why elements and compounds can be classified as: a ionic b simple molecular (covalent) c giant covalent d metallic and how the structure and bonding of these types of substances results in different physical properties, including relative melting point and boiling point, relative solubility in water and ability to conduct electricity (as solids and in solution)	
1.33	Explain the properties of ionic compounds limited to: a high melting points and boiling points, in terms of forces between ions b whether or not they conduct electricity as solids, when molten and in aqueous solution	4a
1.34	Explain the properties of typical covalent, simple molecular compounds limited to: a low melting points and boiling points, in terms of forces between molecules (intermolecular forces) b poor conduction of electricity	4a
1.35	Recall that graphite and diamond are different forms of carbon and that they are examples of giant covalent substances	
1.36	Describe the structures of graphite and diamond	5b
1.37	Explain, in terms of structure and bonding, why graphite is used to make electrodes and as a lubricant, whereas diamond is used in cutting tools	5b
1.38	Explain the properties of fullerenes including C_{60} and graphene in terms of their structures and bonding	5b
1.39	Describe, using poly(ethene) as the example, that simple polymers consist of large molecules containing chains of carbon atoms	5b
1.40	Explain the properties of metals, including malleability and the ability to conduct electricity	5b
1.41	Describe the limitations of particular representations and models, to include dot and cross, ball and stick models and two- and three-dimensional representations	5b
1.42	Describe most metals as shiny solids which have high melting points, high density and are good conductors of electricity whereas most non-metals have low boiling points and are poor conductors of electricity	

Use of mathematics

- Represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon (5b).
- Translate information between diagrammatic and numerical forms (4a).

Calculations involving masses

Students should:		Maths skills
1.43	Calculate: a relative formula mass given relative atomic masses b percentage by mass of an element in a compound given relative atomic masses	1a, 1c
1.44	Calculate the formulae of simple compounds from reacting masses or percentage composition and understand that these are empirical formulae	1a, 1c 2a
1.45	Deduce: a the empirical formula of a compound from the formula of its molecule b the molecular formula of a compound from its empirical formula and its relative molecular mass	1c
1.46	Describe an experiment to determine the empirical formula of a simple compound such as magnesium oxide	1a, 1c 2a
1.47	Explain the law of conservation of mass applied to: a a closed system including a precipitation reaction in a closed flask b a non-enclosed system including a reaction in an open flask that takes in or gives out a gas	1a
1.48	Calculate masses of reactants and products from balanced equations, given the mass of one substance	1a, 1c 2a
1.49	Calculate the concentration of solutions in g dm^{-3}	1a, 1c 2a 3b, 3c
1.50	**Recall that one mole of particles of a substance is defined as:** **a the Avogadro constant number of particles (6.02×10^{23} atoms, molecules, formulae or ions) of that substance** **b a mass of 'relative particle mass' g**	1b

Students should:		Maths skills
1.51	**Calculate the number of:** **a moles of particles of a substance in a given mass of that substance and vice versa** **b particles of a substance in a given number of moles of that substance and vice versa** **c particles of a substance in a given mass of that substance and vice versa**	1a, 1b, 1c 2a 3a, 3b, 3c
1.52	**Explain why, in a reaction, the mass of product formed is controlled by the mass of the reactant which is not in excess**	1c
1.53	**Deduce the stoichiometry of a reaction from the masses of the reactants and products**	1a, 1c

Use of mathematics

- Arithmetic computation and ratio when determining empirical formulae, balancing equations (1a and 1c).
- Arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry (1a, 1c and 1d).
- **Calculations with numbers written in standard form when using the Avogadro constant (1b)**.
- Change the subject of a mathematical equation (3b and 3c).
- Provide answers to an appropriate number of significant figures (2a).
- **Convert units where appropriate particularly from mass to moles (1c)**.

Suggested practicals

- Investigate the size of an oil molecule.
- Investigate the properties of a metal, such as electrical conductivity.
- Investigate the different types of bonding: metallic, covalent and ionic.
- Investigate the typical properties of simple and giant covalent compounds and ionic compounds.
- Classify different types of elements and compounds by investigating their melting points and boiling points, solubility in water and electrical conductivity (as solids and in solution), including sodium chloride, magnesium sulfate, hexane, liquid paraffin, silicon(IV) oxide, copper sulfate, and sucrose (sugar).
- Determine the empirical formula of a simple compound.
- Investigate mass changes before and after reactions.
- Determine the formula of a hydrated salt such as copper sulfate by heating to drive off water of crystallisation.